

Game-Based Learning for Classical Chinese: A Theory-Driven Narrative Design in the Age of AI

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Introduction

Game-Based Learning (GBL) uses an immersive, interactive, and student-centered nature to enhance learning experiences and create engaging environments.

Practical Needs: 1. GBL's digital narratives offer diverse story branches, unlike traditional linear texts. This approach encourages players to reflect deeply and understand themes by making choices.

2. AI-generated images can help learners understand abstract Classical Chinese concepts, presenting abstract concepts and scenarios in a concrete setting. Moreover, AI simplifies the process of visualizing difficult ideas and significantly reduces challenges in instructional design.



Figure 1. Falling petal effect.

Research Gaps: GBL research in China has largely concentrated on STEM and EFL fields, with limited applications in literature and humanities education and promotion.

Research Aims

- 1.Provided actionable technical solutions and guidelines for gamifying classical Chinese texts.
- 2.To generate a generalizable Narrative Exploration Gamification Design Framework and then to examine the framework's general applicability, flexibility, and universality (In progress).

Theoretical Framework

Constructivist Learning Theory: *How learners actively construct knowledge?*

The game's scene construction provides an "situated learning" environment. Within this context, students act as active participants, constructing their understanding of Classical Chinese through immersive gameplay.

Self-Determination Theory: *Why students want to learn in the game environment?*

GBL fosters intrinsic motivation by satisfying three psychological needs: Autonomy (agency in decision-making), Competence (overcoming challenges with instant feedback), and Relatedness (emotional connection to the storytelling).

Serious Game: *Providing an adaptation theory to apply the literary materials into gamification.*

It prioritizes pedagogical outcomes over entertainment, and uses narrative and interaction to meet the goals of SDT and CLT.

Diegesis Theory: *How to balance the game tasks and plots design?*

It combines "intra-diegetic elements" (like characters) for immersion and "extra-diegetic elements" (like external resources) for learning.

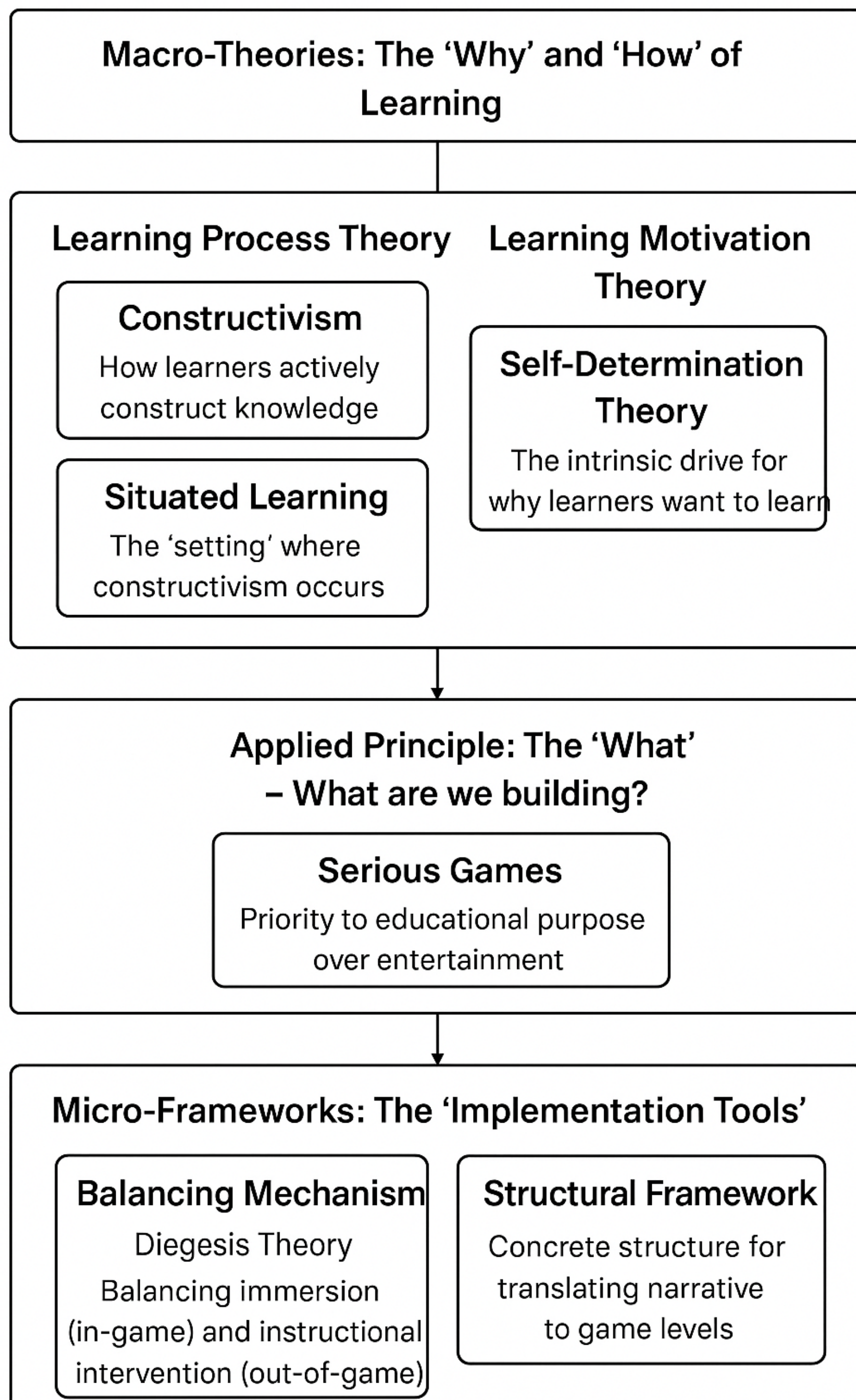


Figure 2. This is a theoretical map.

Narrative Configuration & Challenges

Interaction & Learning Design

1.Players interact and progress the storyline using only mouse clicks, which helps to minimize cognitive load. The game retains the original content. It also adds plots and new endings through character dialogues and decision-making mechanism.

2.Problem:Teaching prompts (like background knowledge) may cause excessive cognitive load and disrupt immersion. Solution:Break down the teaching tasks and introduce prompts after branch decisions (rather than at key plot points).

Multi-Ending Design

1.Multi-ending encourages players to reflect deeply on the text's themes and to explore contrasting viewpoints by making their own personal choices.

2.Challenges:Multiple endings can compromise the text's overall integrity. They may increase students' cognitive load and provide too much freedom in dialogue choices, which may lead students to miss essential knowledge points and weaken the text's main theme.

3.Solutions:To address these challenges, the main storyline can be designed as the central narrative. Essential knowledge may be embedded at "mandatory narrative checkpoints" to prevent students from missing key teaching objectives. At the same time, transforming complex text into "interrogative dialogues" and providing a "back" function helps maintain narrative clarity and instructional focus.

Endings	Main idea of the endings
The Vanishing Path (不复得路)	Tragic: Can human desires and the utopian society coexist?
Ephemeral Kingdom (南柯一梦)	Fantastical:Does the utopian society only exist in imagination?
Awakening in Liminal Rain (雨夜梦醒)	Conflicting: Does external interference threaten the delicate harmony of resources in the utopian society?
Invisible in the Crowd (大隐隐于世)	Compromising: A balance between the dream of seclusion and the demands of real life.
Timeless Sanctuary (不知有汉)	Fantastical:The dislocation of time and space creates a mirage-like illusion of the utopian society.
The Seeker from the Future (南阳刘子骥)	Tragic:The conflict between personal aspirations and societal opinions.

Figure 3. Multi-ending design.

Applying the Diamond Narrative Model

1.This model divides the text into six parts, for adapting into game tasks or plots. 2.Interactive effects and music build an immersive scene in background section. The action section inserts exploration tasks to embed historical knowledge.

Narrative Exploration Gamification Design Framework

Primary Indicator	Secondary Indicator (Sub-feature)	Scoring Description (Score of 2)
Scene Construction	1a. Physical Space Progression	The text provides a rich, clear, and explorable physical environment and elements.
	1b. Dialectical Reflective Scenario	The text builds a clear "game board" of situational choices, power dynamics, or moral dilemmas.
Narrative Conflict	2a. External Plot Conflict	The text contains clear character opposition or event-driven conflicts that are directly to convert into tasks.
	2b. Internal Speculative Conflict	The text's core is a deep internal struggle, value judgment, or philosophical debate.
Player Agency & Perspective	3a. Suitability of Primary Perspective	Playing as the text's narrator offers a highly immersive and emotionally resonant experience.
Pedagogical Transformation	4a. Potential for Symbolic Objectives	The text is rich with specific, concrete objects or images that can be made interactive (e.g., a specific pavilion, a type of plant, a weapon).
	4b. Potential for Embedded Knowledge	The text's historical context, philosophical ideas, or literary devices are suitable for being explained through supplementary, interactive elements (e.g., tooltips, pop-ups).

Figure 4. Narrative Exploration Gamification Design Framework.

The NEGDF contains four elements:

Scene Construction serves as providing immersive and concrete environment.

Narrative Conflicts offer diverse narrative branches, leading to different endings.

Player Agency & Perspective provides a substitutable perspective and predefines an identity stance for the player, laying the groundwork for the subsequent plot conflicts.

Pedagogical Transformation means that insert pedagogical elements, like knowledge points in pop-up windows (figure 5). It supplements background knowledge and integrates points into students' knowledge construction.



Figure 5. Pop-up windows .

Conclusion

This research formulated a NEGDF and proposed actionable gamification guidelines. Future work will use a qualitative matrix to examine the framework's general applicability, flexibility, and universality in other texts, alongside more extensive empirical validation.